

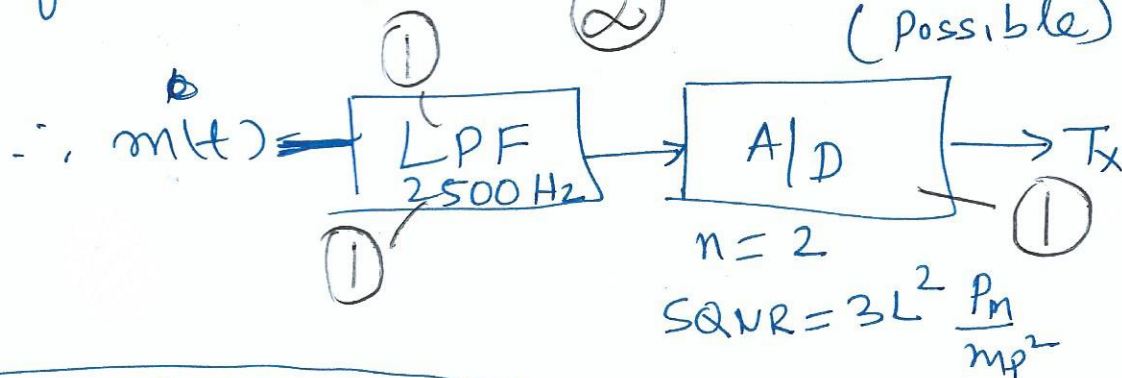
Q.1 FS of $m(t)$ (Inspired by ring modulator)

$$m(t) = \frac{4}{\pi} \left[\cos 1000\pi t - \frac{1}{3} \cos 3000\pi t + \frac{1}{5} \cos 5000\pi t - \dots \right]$$

(a) $BW = 5 \text{ KHz}$, $R_b = 10 \text{ Kbps}$ (2)

If $n=1$ $f_s = 10 \text{ Ksamples/s}$, $f_m = 5 \text{ KHz}$
(not possible)

If $n=2$ $f_s = 5 \text{ Ksamples/s}$, $f_m = 2500 \text{ Hz}$
(possible)

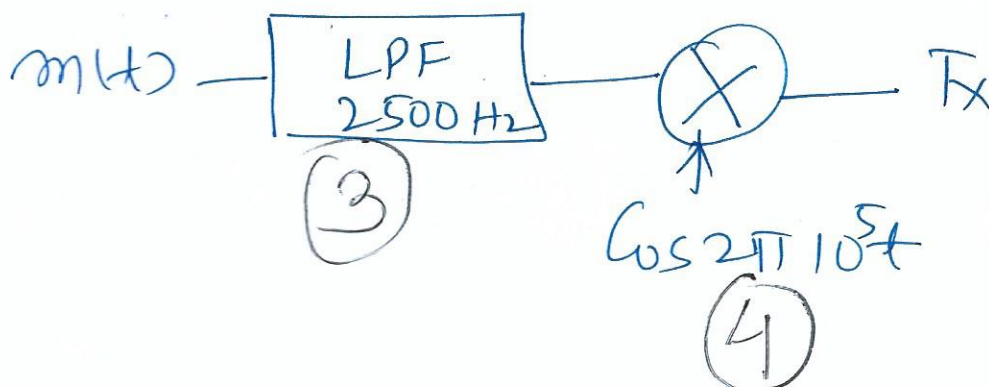


$$\text{SNR} = 3L^2 \frac{P_m}{m_p^2}$$

$$= 3(16) \frac{(1)}{(1)^2} = 48$$

Multiple approaches are possible for this design problem

(b) $BW = 5 \text{ KHz} = \text{channel BW}$
DSB-SC $BW = 5 \text{ KHz}$
Message BW $B_{\text{message}} = 2500 \text{ Hz}$



Q.1
(c)

$$\beta = 4$$

$$\boxed{\begin{aligned} 2(\Delta f + f_m) &= 5000 \\ \Delta f + f_m &= 2500 \end{aligned}}$$

$$\frac{\Delta f}{f_m} = 4 \Rightarrow \frac{2500 - f_m}{f_m} = 4$$

$$\boxed{\therefore f_m = 500 \text{ Hz}} \quad \boxed{\Delta f = 2000 \text{ Hz}}$$

$$m(t) \rightarrow \boxed{\text{LPF } 500 \text{ Hz}} \rightarrow \frac{4}{\pi} \cos 1000\pi t = \boxed{\text{FM}} \rightarrow x(t)$$

$$K_f m_p' = 2000 \Rightarrow \boxed{K_f = 500 \pi \text{ Hz/Volt}}$$

$$x(t) = \cos(2\pi 10^5 t + 4000\pi \int_0^t \cos 1000\pi z dz)$$

$$x(t) = \cos(2\pi 10^5 t + 4 \sin 1000\pi t)$$

$$= \sum_{n=-\infty}^{\infty} J_n(\beta) \cos(2\pi(10^5 + 500n)t)$$

$$m(t) \rightarrow \boxed{\text{LPF } 500 \text{ Hz}} \rightarrow \frac{4}{\pi} \cos 1000\pi t \rightarrow \boxed{\text{FM Mod.}} \rightarrow \boxed{\text{BPF}} \rightarrow$$

$$\textcircled{2} \quad f_c = 100 \text{ kHz} \\ \text{BW} = 5 \text{ kHz}$$

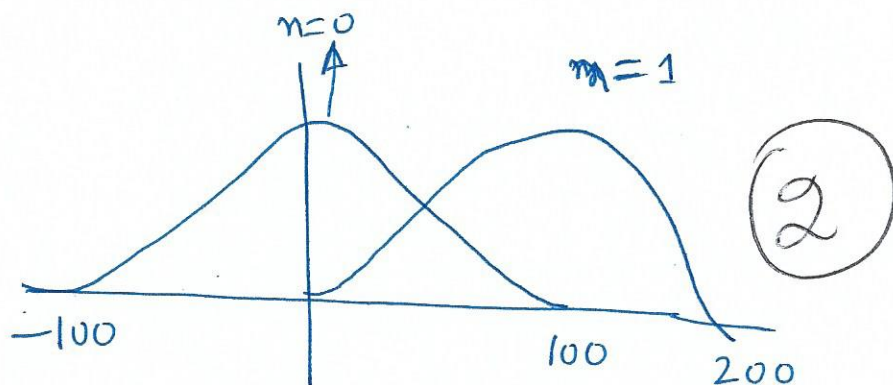
$$\textcircled{2} \\ f_m = 500 \text{ Hz}$$

①

$$\textcircled{1} - K_f = 500 \pi \text{ Hz/Volt} \\ f_c = 100 \text{ kHz}$$

$$\textcircled{1} - \Delta f = 2000 \text{ Hz}$$

Q. 2
= (a)



$$\rightarrow R_s = 100$$

$$\sum_{n=-\infty}^{\infty} p(t + nR_s) = \text{Constant}$$

$$\therefore \boxed{R_s = 100}$$

$$BW = 100$$

$$BW = \frac{(\alpha + 1) R_s}{2}$$

$$\therefore \boxed{\alpha = 1} \text{ Full raised cosine Pulse.}$$

$$\therefore p(t) = R_s \left(\frac{\cos \pi R_s t}{1 - 4 R_s^2 t^2} \right) \text{Sinc}(\pi R_s t)$$

$$\boxed{p(t) = 100 \left(\frac{\cos \pi 100 t}{1 - 40000 t^2} \right) \text{Sinc}(100 \pi t)}$$

Or,

$$p(t) = \frac{\cos(\pi 100 t)}{1 - 40000 t^2} \text{Sinc}(100 \pi t)$$

Q 2 (b) Signal power

Amplitude of signal at the output
of matched filter = Energy of $p(t)$

$$p(t) = t ; \quad t = 0 \text{ to } 1$$

$$= 1 ; \quad t = 1 \text{ to } 3$$

$$= -t+4 ; \quad t = 3 \text{ to } 4$$

$$\therefore E_p = \int_0^1 t^2 dt + \int_1^3 1^2 dt + \int_3^4 (-t+4)^2 dt$$

$$= \frac{1}{3} + 2 + \frac{1}{3} = \frac{8}{3}$$

$$\therefore \text{Amplitude} = \frac{8}{3}$$

$$\boxed{\text{Signal power} = \frac{64}{9} = 7.11} \quad (5)$$

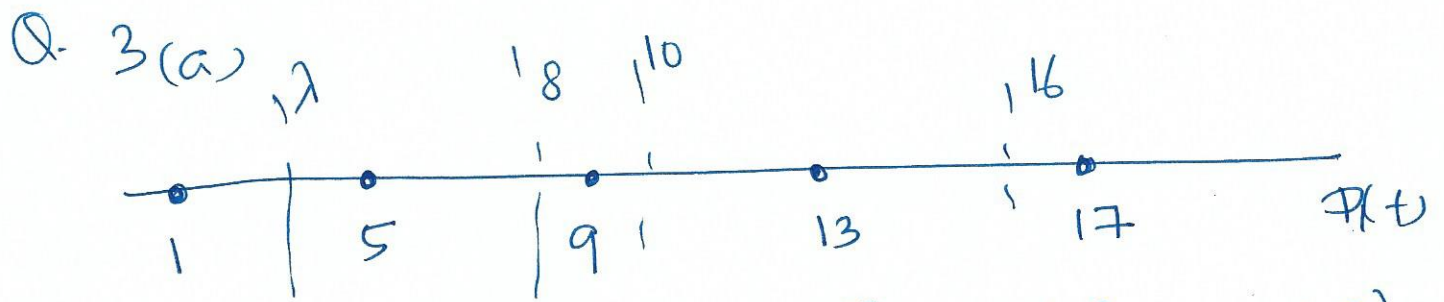
Noise power at the o/p of MF :-

$$P_{no} = \int_{-\infty}^{\infty} S_n(f) (F(f))^2 df = \frac{N_0}{2} \int_{-\infty}^{\infty} (F(f))^2 df$$

$$= \frac{N_0}{2} \int_{-\infty}^{\infty} |f(t)|^2 dt = \frac{N_0}{2} E_p = \frac{4N_0}{3}$$

$$\therefore P_{no} = \frac{4N_0}{3} = 1.33N_0 \quad (5)$$

$$\boxed{\text{BER} = Q \sqrt{\frac{2E_p}{N_0}} = Q \sqrt{\frac{16}{3N_0}} = Q \sqrt{\frac{5.33}{N_0}}} \quad (4)$$



$$P_c = P_1 P(1+n < \lambda) + P_2 P[n+5 > \lambda, n+5 > 8] \\ + P_3 P[9+n > 8, 9+n < 10] + P_4 P[n+13 > 10, n+13 < 16] \\ + P_5 P[17+n > 16] \quad (4)$$

$$P_c = P_1 P[n < \lambda - 1] + P_2 P[\lambda - 5 < n < 3] \\ + P_3 P(-1 < n < 1) + P_4 P(3 < n < 3) \\ + P_5 P(n > -1) \quad (\text{scribble})$$

$$P_c = P_1 Q(1-\lambda) + P_2 Q(\lambda-5) - Q(3) \\ + P_3 (1 - 2Q(1)) + P_4 (1 - 2Q(3)) \\ + P_5 (1 - Q(1)) \quad (2)$$

$$P_e = 1 - P_c$$

$$\frac{dP_e}{d\lambda} = - \left[P_1 (-1) \frac{1}{\sqrt{2\pi}} e^{-\frac{(1-\lambda)^2}{2}} (-1) + P_2 (-1) \frac{1}{\sqrt{2\pi}} e^{-\frac{(\lambda-5)^2}{2}} \right] = 0 \quad (2)$$

$$P_1 e^{-\frac{(1-\lambda)^2}{2}} = P_2 e^{-\frac{(\lambda-5)^2}{2}}$$

$$\lambda = 3 + \frac{1}{4} \log \frac{P_1}{P_2} \quad (4)$$

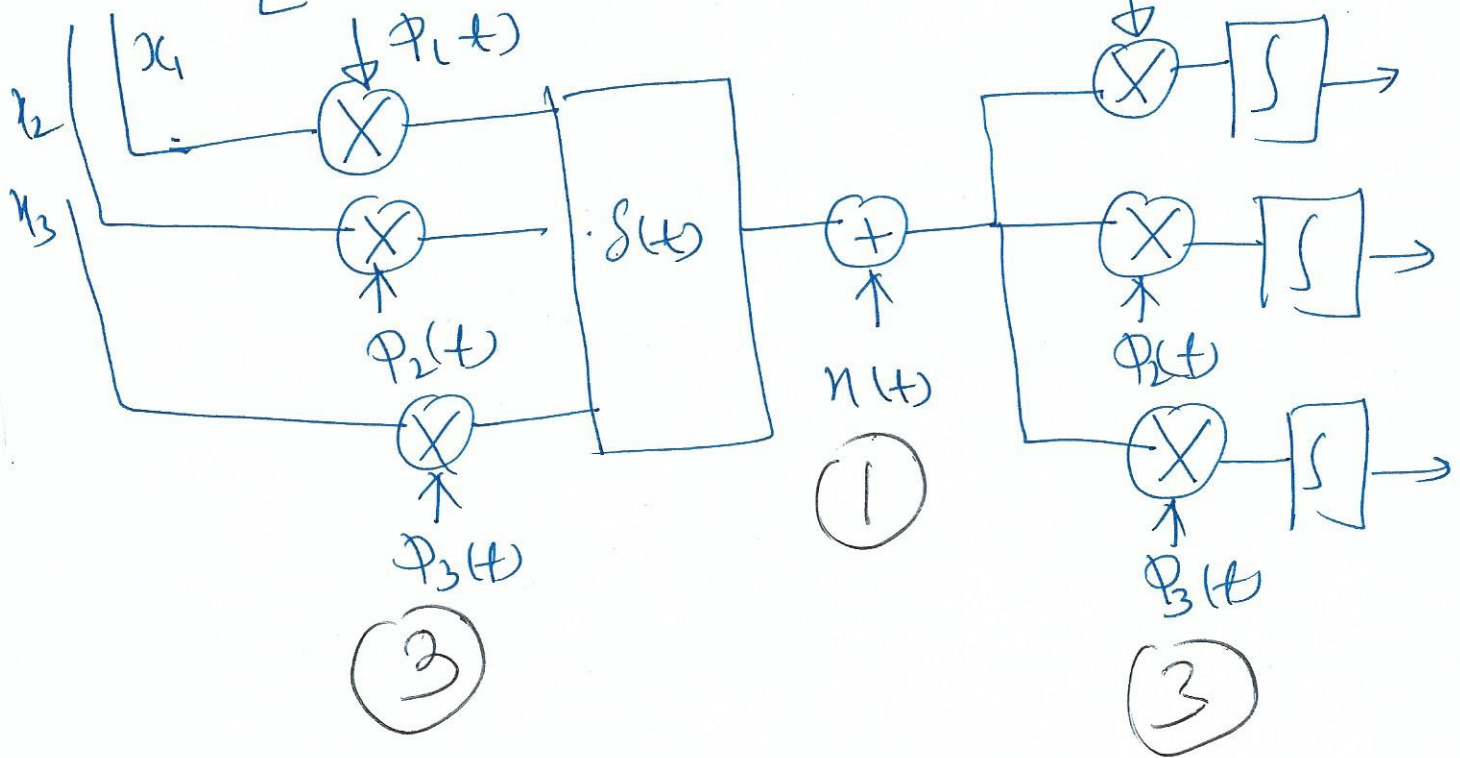
$$P_e = 1 - P_c \text{ at } \lambda = 3 + \frac{1}{4} \log \frac{P_1}{P_2} \quad (2)$$

Q.3(b)

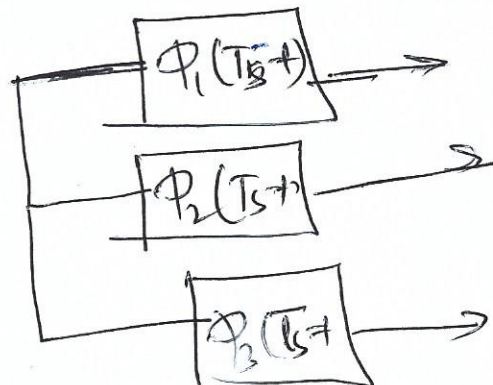
Constellation

$$X = \{(a, 0, 0), (0, b, 0), (0, 0, c)\}$$

MF (Correlator)

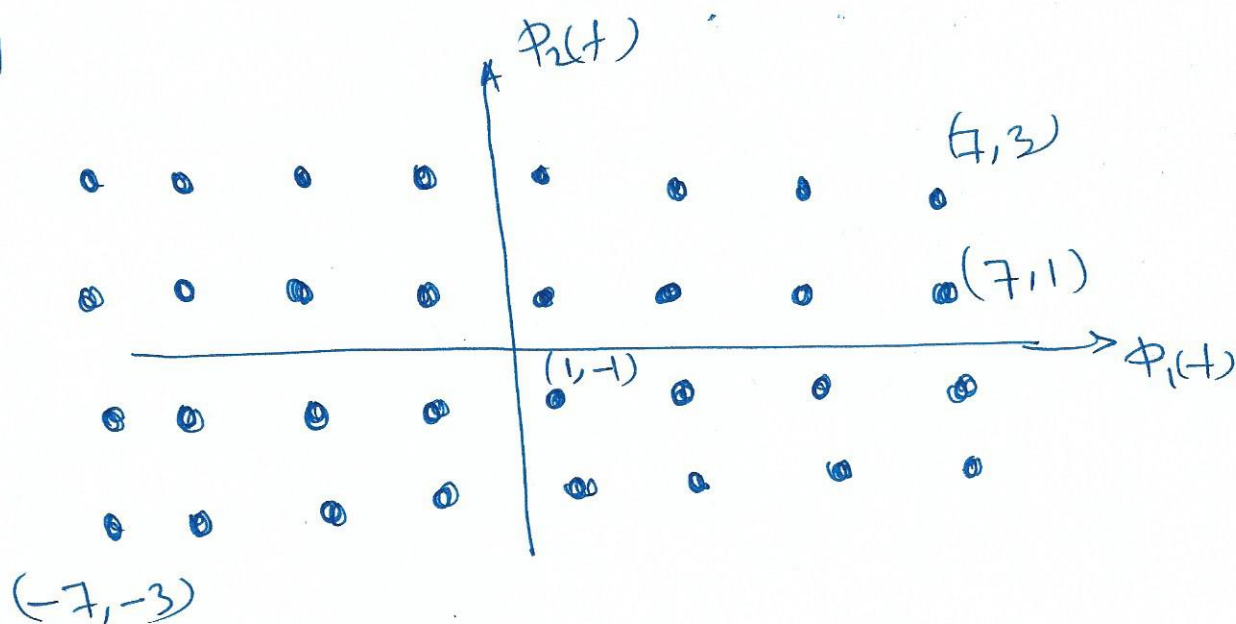


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MF (Filter)

Q.4



$$(a) P_c = P[-7 + \eta_1 < -6, -3 + \eta_2 < -2]$$

$$P_c = P[\eta_1 < 1, \eta_2 < 1]$$

$$P_c = \left[1 - Q\left(\frac{1}{\sigma_{\eta_1}}\right)\right] \left[1 - Q\left(\frac{1}{\sigma_{\eta_2}}\right)\right]$$

$$P_c = 1 - Q\left(\frac{1}{2}\right) - Q\left(\frac{1}{4}\right) - Q\left(\frac{1}{2}\right)Q\left(\frac{1}{4}\right) \quad (5)$$

$$(b) P_c = P[\eta_1 + 1 < 2, \eta_1 + 1 > 0; \eta_2 - 1 < 0, \eta_2 - 1 > -2]$$

$$P_c = P[1 + \eta_1 < 2, 1 + \eta_1 > 0; -1 + \eta_2 < 0, -1 + \eta_2 > -2]$$

$$= P[-1 < \eta_1 < 1; -1 < \eta_2 < 1]$$

$$= \left[1 - 2Q\left(\frac{1}{2}\right)\right] \left[1 - 2Q\left(\frac{1}{4}\right)\right]$$

$$P_c = 1 + 4Q\left(\frac{1}{2}\right)Q\left(\frac{1}{4}\right) - 2Q\left(\frac{1}{2}\right) - 2Q\left(\frac{1}{4}\right) \quad (5)$$

(C)

First Symbol in the first quadrant
(outer most)

$$P_4 = P[(3+n_2 > 2)] = P[n_2 > -1]$$
$$= \int_{-1}^{\infty} \frac{1}{2} e^{-x} dx = \int_{-1}^0 \frac{1}{2} e^x dx + \int_0^{\infty} \frac{1}{2} e^{-x} dx$$

$$P_{c1} = \frac{1}{2} [1 - e^{-1}] + \frac{1}{2} = 1 - \frac{e^{-1}}{2}$$

First row and Fourth row $\Rightarrow$ All symbols will have

$$P_{c1} \dots = P_{c4} = 1 - \frac{e^{-1}}{2} \quad (4)$$

2nd row, first quadrant, outer most

$$P_{c1}^1 = P[1+n_2 > 0, 1+n_2 < 2]$$

$$= P(-1 < n_2 < 1)$$

$$= \int_{-1}^1 \frac{1}{2} e^{-x} dx = \int_{-1}^0 \frac{1}{2} e^{+x} + \int_0^1 \frac{1}{2} e^{-x} dx$$

$$= \frac{1}{2} [1 - e^{-1}] + \frac{1}{2} [1 - e^{-1}] = 1 - e^{-1}$$

2nd row & Third row = All symbols have

$$P_{c1}^1 = \dots = P_{c4}^1 = 1 - e^{-1} \quad (4)$$

$$\therefore P_c = \frac{1}{2} [1 - \frac{e^{-1}}{2} + 1 - e^{-1}] = 1 - \frac{3}{4} e^{-1}$$

$$\therefore P_c = \frac{3}{4} e^{-1} = \frac{3}{4e} \quad (3)$$

Q.5 (a)

Transition Matrix

$$\begin{array}{c} X_1 \\ X_2 \end{array} \begin{array}{ccc} Y_1 & Y_2 & Y_3 \\ \left[\begin{array}{ccc} 0.6 & 0.3 & 0.1 \\ 0.1 & 0.3 & 0.6 \end{array} \right] \end{array}$$

$$P(X_1) = P(X_2) = \frac{1}{2} \Rightarrow H(X) = 1 \text{ bits/symbol}$$

$$P(Y_1) = P(Y_1|X_1)P(X_1) + P(Y_1|X_2)P(X_2) \\ = 0.35$$

$$P(Y_2) = 0.3 \quad P(Y_3) = 0.35$$

$$H(Y) = 1.5723 \text{ bits/symbol} \quad (3)$$

$$H(Y|X) = \sum_{j=1}^3 \sum_{i=1}^2 P(Y_j, X_i) \frac{1}{\log_2(P_j|X_i)}$$

$$= 1.2865 \text{ bits/symbol} \quad (3)$$

$$C = H(Y) - H(Y|X) = 1.5723 - 1.2865$$

$$C_s = 0.2858 \text{ bits/symbol} \quad (1)$$

Alternatively $H(X) = 1$; $H(X|Y) = \sum_{i=1}^2 \sum_{j=1}^3 P(X_i, Y_j) \frac{1}{\log_2(P_i|Y_j)}$

$$= 0.7142 \quad (5)$$

$$C_s = H(X) - H(X|Y) = 0.2858 \text{ bits/symbol} \quad (1)$$

Q.5 (b) $C_s = H(Y) - H(V)$ bits/symbol

for $N(0,1) \Rightarrow H_1 = \frac{1}{2} \log_2 (2\pi e \sigma^2)$
 $= 2.044$ ①

for $N(\mu, \sigma^2) \Rightarrow H_2 = \int_{-\infty}^{\infty} f(x) \log_2 \sqrt{2\pi\sigma^2} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$
 $= \int_{-\infty}^{\infty} f(x) \log_2 (2\pi\sigma^2)^{\frac{1}{2}} + \int_{-\infty}^{\infty} (\log_2 e) \frac{(x-\mu)^2}{2\sigma^2} f(x) dx$
 $= \frac{1}{2} \log_2 (2\pi\sigma^2) + (\log_2 e) * \frac{\sigma^2}{2\sigma^2}$
 $= \frac{1}{2} \log_2 (2\pi\sigma^2) + \frac{1}{2} \log_2 e$
 $= \frac{1}{2} \log_2 (2\pi e \sigma^2)$

$\therefore N(1,2) \Rightarrow H_2 = \frac{1}{2} \log_2 (2\pi(2.7)(2))$
 $= 2.54$ ②

for $U(-1,1) \Rightarrow H_3 = \log_2 2 = 1$ ①

Maximum capacity $C_s = 2.54 - 1 = 1.54$ bits/symbol ①

Minimum capacity $C_s = 2.54 - 2.04$
 $= 0.5$ bits/symbol ①

Or
 Minimum capacity = 0

Q.5 (c)

$$C_1 = \begin{matrix} 1 \times 4 \\ 4 \times 7 \end{matrix} [0000] \begin{matrix} 7 \\ 4 \end{matrix} = 00000000$$

$$C_2 = [0001] \begin{matrix} 7 \\ 4 \end{matrix} = 0001101$$

$$C_3 = [0010] \begin{matrix} 7 \\ 4 \end{matrix} = 0011010$$

$$C_4 = [0011] \begin{matrix} 7 \\ 4 \end{matrix} = 001011$$

$$C_5 = [0100] \begin{matrix} 7 \\ 4 \end{matrix} = 0110100$$

$$C_6 = [0101] \begin{matrix} 7 \\ 4 \end{matrix} = 0111001$$

$$C_7 = [0110] \begin{matrix} 7 \\ 4 \end{matrix} = 0101110$$

$$C_8 = [0111] \begin{matrix} 7 \\ 4 \end{matrix} = 0100011$$

$$C_9 = [1000] \begin{matrix} 7 \\ 4 \end{matrix} = 1101000$$

$$C_{10} = [1001] \begin{matrix} 7 \\ 4 \end{matrix} = 1100101$$

$$C_{11} = [1010] \begin{matrix} 7 \\ 4 \end{matrix} = 1110010$$

$$C_{12} = [1011] \begin{matrix} 7 \\ 4 \end{matrix} = 1111111$$

$$C_{13} = [1100] \begin{matrix} 7 \\ 4 \end{matrix} = 1011100$$

$$C_{14} = [1101] \begin{matrix} 7 \\ 4 \end{matrix} = 1010001$$

$$C_{15} = [1110] \begin{matrix} 7 \\ 4 \end{matrix} = 1000110$$

$d_{\min} = 3$
 $t+1 = 3$
 $t = 2$
1

~~5~~

6

$$C_{16} = [1111] \begin{matrix} 7 \\ 4 \end{matrix} = 1001011$$